

Project Report On

2D Tennis Game

Project - I
Course code: 1200

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CERTIFICATE

This is to certify that the project report entitled “Development of a Simple Tennis Game using C++ and SFML” submitted by MD. Imran Hosen to the Department of Information and Communication Technology, Mawlana Bhashani Science and Technology University has been carried out under my supervision and guidance. It is further certified that this project has not been submitted elsewhere for the award of any degree or diploma.

Signature of Supervisor

Mr. Mahedi Hasan

Lecturer,

Department of ICT, MBSTU

DECLARATION

I hereby declare that the project report entitled “Development of a Simple Tennis Game using C++ and SFML” is an original work carried out by me under the supervision of Mr. Mahedi Hasan sir. This report has not been submitted previously to any institution for any academic qualification or diploma.

Signature of Student

MD. Imran Hosen

ACKNOWLEDGEMENT

First and foremost, I thank Almighty Allah for granting me the wisdom, health, and patience to complete this project. I would like to express my sincere gratitude to my supervisor, Mr. Mahedi Hasan, for his valuable guidance, constant encouragement, and insightful suggestions throughout the development of this project. I also extend my appreciation to the Department of Information and Communication Technology, Mawlana Bhashani Science and Technology University, for providing the necessary facilities and resources. My heartfelt thanks to my classmates, friends, and family for their continuous support and motivation during the project work.

ABSTRACT

This project focuses on the development of a simple 2D Tennis Game using C++ and the SFML (Simple and Fast Multimedia Library). The primary objective of this project was to gain hands-on experience in game programming concepts such as event handling, real-time rendering, and collision detection. The project successfully implements a playable two-player game where each player controls a paddle to hit a moving ball, and scores are displayed dynamically. The system demonstrates the application of object-oriented programming, game loops, and physics simulation. In the future, this project can be extended by adding artificial intelligence (AI) opponents, sound effects, and advanced graphics.

INTRODUCTION

The gaming industry is one of the fastest-growing fields in computer science. Developing simple games serves as a foundation for learning essential programming concepts such as graphics rendering, user input, and real-time updates. This project presents the development of a simple Tennis Game inspired by the classic Pong game, using C++ and SFML.

The main objective is to understand and implement the logic behind a two-player interactive game. Players can move paddles up and down using keyboard keys, while a ball bounces between them. The player who fails to return the ball loses a point.

The system is divided into different layers including input handling, physics and collision logic, and graphics rendering. The project offers a strong foundation for understanding event-driven programming and real-time system design.

SYSTEM DESIGN

The development of the Tennis Game followed a structured methodology to ensure modularity and simplicity.

The system architecture consists of the following layers:

1. Input Layer – Handles keyboard inputs from players for paddle movement.
2. Game Logic Layer – Updates the ball's position, checks collisions, and manages scores.
3. Rendering Layer – Draws paddles, ball, and score on the screen using SFML functions.

Tools and Technologies Used:

- Programming Language: C++
- Graphics Library: SFML 3.0
- IDE: Visual Studio Code

The main loop of the game includes event polling, updating objects, and rendering frames continuously to achieve smooth gameplay.

TESTING

Testing was conducted at various stages to ensure that each game component performed as expected.

1. Unit Testing: Each class and function, including paddle movement and ball collision detection, was tested individually.
2. Integration Testing: Verified smooth interaction between ball and paddle, and ensured score updates correctly.
3. System Testing: Checked the game performance, frame rate stability, and user input responsiveness.

Test Cases Included:

- Ball bouncing off walls and paddles.
- Correct score increment after each miss.
- Game reset functionality after reaching score limit.

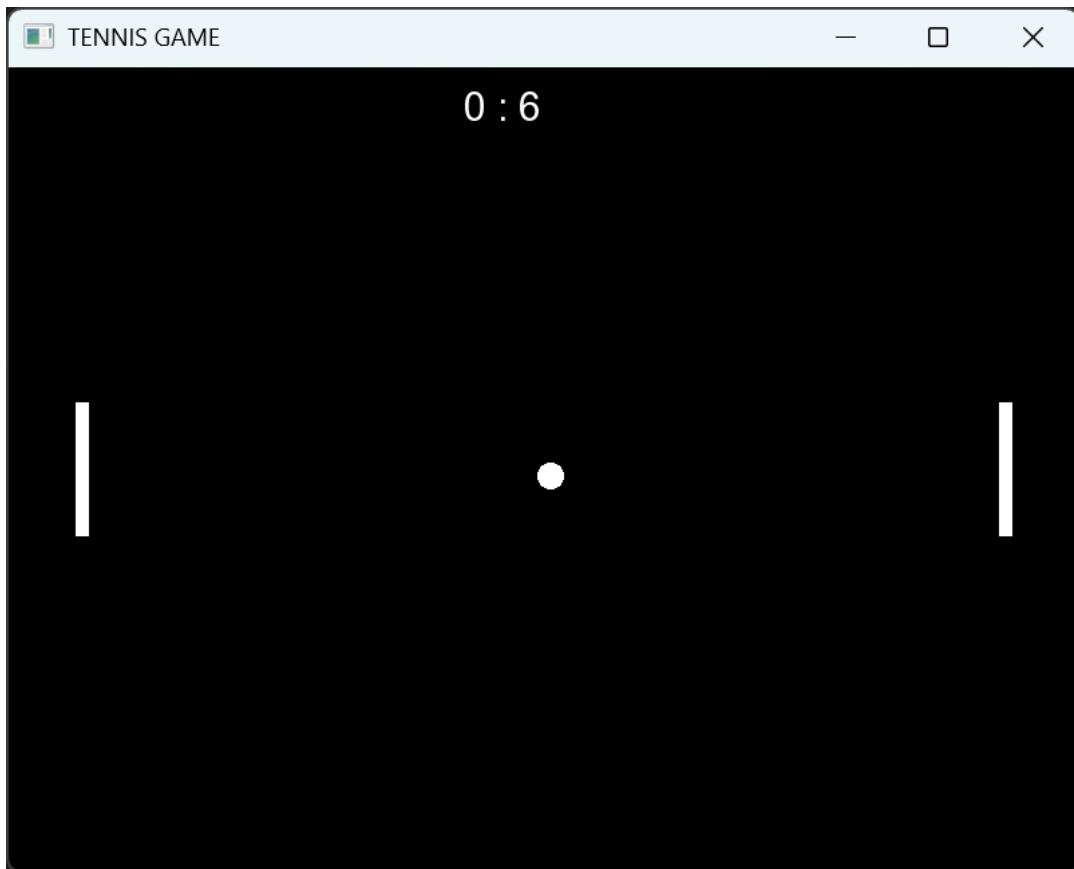
The results confirmed that the Tennis Game runs smoothly without runtime errors and accurately tracks scores.

CONCLUSION

The Tennis Game project successfully demonstrated the fundamentals of game development using C++ and SFML. It achieved all its objectives, including smooth paddle movement, collision detection, score tracking, and real-time rendering. The project helped enhance programming skills in handling graphics, events, and object-oriented design.

Future enhancements could include single-player AI opponents, dynamic difficulty levels, sound effects, and improved visual aesthetics. This project represents an important step toward mastering interactive software development.

Project Demo



This image shows the main window of the Tennis Game built using SFML. It initializes the background, paddles, and ball. The paddles are placed at the top and bottom, and the game title “TENNIS GAME” appears on the window bar

SOURCE CODE

```
#include <SFML/Graphics.hpp>

#include <string>

using namespace sf;

int main() {

    RenderWindow window(VideoMode({800, 600}), "TENNIS GAME");

    window.setFramerateLimit(60);


    // Bat

    RectangleShape paddle1({10.f, 100.f});

    paddle1.setPosition({50.f, 250.f});

    paddle1.setFillColor(Color::White);


    RectangleShape paddle2({10.f, 100.f});

    paddle2.setPosition({740.f, 250.f});

    paddle2.setFillColor(Color::White);


    CircleShape ball(10.f);

    ball.setPosition({395.f, 295.f});

    ball.setFillColor(Color::White);


    // velocity
```

```

Vector2f ballVelocity{-4.f, -3.f};

float paddleSpeed = 5.f;


// Score

int score1 = 0;

int score2 = 0;


Font font;

if (!font.openFromFile("resources/arial.ttf")) {
    return -1;
}


Text scoreText(font, "0 - 0", 30);

scoreText.setFillColor(Color::White);

scoreText.setPosition({340.f, 10.f});


while (window.isOpen()) {
    while (auto e = window.pollEvent()) {
        if (e->is<Event::Closed>())
            window.close();
    }
}

if(score1<10 && score2<10){

    // Player 1

```

```
    if (Keyboard::isKeyPressed(Keyboard::Scan::W) && paddle1.getPosition().y > 0)
```

```
        paddle1.move({0.f, -paddleSpeed});
```

```
    if (Keyboard::isKeyPressed(Keyboard::Scan::S) && paddle1.getPosition().y < 500)
```

```
        paddle1.move({0.f, paddleSpeed});
```

```
// Player 2
```

```
    if (Keyboard::isKeyPressed(Keyboard::Scan::Up) && paddle2.getPosition().y > 0)
```

```
        paddle2.move({0.f, -paddleSpeed});
```

```
    if (Keyboard::isKeyPressed(Keyboard::Scan::Down) && paddle2.getPosition().y < 500)
```

```
        paddle2.move({0.f, paddleSpeed});
```

```
    ball.move(ballVelocity);
```

```
    if (ball.getPosition().y + ball.getRadius() <= 0 || ball.getPosition().y + ball.getRadius()*2 >= 600)
```

```
        ballVelocity.y *= -1.f;
```

```
FloatRect ballBounds = ball.getGlobalBounds();
```

```
FloatRect paddle1Bounds = paddle1.getGlobalBounds();
```

```
FloatRect paddle2Bounds = paddle2.getGlobalBounds();
```

```
    if (ballBounds.findIntersection(paddle1Bounds) ||
ballBounds.findIntersection(paddle2Bounds)) {
        ballVelocity.x *= -1.f;
    }
```

```
if (ball.getPosition().x < 0) {
    score2++;
    ball.setPosition({395.f, 295.f});
    ballVelocity = {-4.f, -3.f};
}
```

```
if (ball.getPosition().x > 800) {
    score1++;
    ball.setPosition({395.f, 295.f});
    ballVelocity = {4.f, 3.f};
}
```

```
scoreText.setString(std::to_string(score1) + " : " + std::to_string(score2));
```

```
window.clear(Color::Black);
```

```
window.draw(paddle1);
```

```
window.draw(paddle2);
```

```
window.draw(ball);
```

```
        window.draw(scoreText);
        window.display();
    }
else{
    Text ext(font, "PLAYER WIN", 50);
    ext.setFillColor(Color::White);
    ext.setPosition({240.f, 270.f});
    int sco=score1>score2?1:2;
    ext.setString("PLAYER " + std::to_string(sco)+" WIN");
    window.clear(Color::Black);
    window.draw(ext);
        window.display();
    }
}

return 0; }
```


BIBLIOGRAPHY & REFERENCES

- [1] SFML Documentation – <https://www.sfml-dev.org/>
- [2] Cplusplus.com – C++ Reference, <https://cplusplus.com/>
- [3] Tennis or Pong Game History – <https://en.wikipedia.org/wiki/Pong>
- [4] Stack Overflow Discussions (2024). Collision Detection in 2D Games.
- [5] TutorialsPoint (2024). C++ Game Development using SFML.